

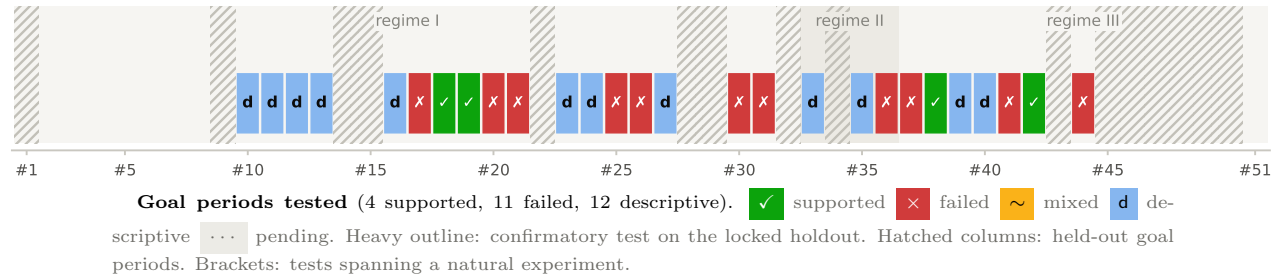
H31: Consensus time scales with the interaction graph's spectral gap

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

not yet rated

Bottom line. Pending.

level: hypothesis · pipeline checks 71% · updated —



WHAT HAPPENED

Not yet summarized. The question: Time for a project or stance to reach a dominant share scales as $1/\lambda_2$ of the exposure Laplacian (the HH68 idea), or as N^a . *Check:* consensus times across periods vs λ_2 and N ; predict held-out periods. Practical aim (usefulness-first batch): produce a number or rule an operator of an LLM agent swarm could compute from logs and act on.

PHYSICAL MODEL

From: 10, graph diffusion. The round-1 agent specifies the exact variant.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Observables. Pending.

RESULTS VS PREDICTIONS

Pending.

GOAL PERIODS IN DETAIL

- G10 d descriptive
 - G11 d descriptive
 - G12 d descriptive
 - G13 d descriptive
 - G16 d descriptive
 - G17 × failed
 - G18 ✓ supported
 - G19 ✓ supported
 - G20 × failed
 - G21 × failed
 - G23 d descriptive
 - G24 d descriptive
 - G25 × failed
 - G26 × failed
 - G27 d descriptive
 - G30 × failed
- +11 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	–	F identifiability	–
B assumptions	–	G ground truth	–
C adequacy	–	H comparative	–
D unfitted predictions	–	I transfer	–
E interventional	–		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Pending.

ROUND 2

No round-2 redirects yet.